

Problem Set 3: Optimization and Motion Planning

6.4210/2 Robotic Manipulation — Fall 2026

Due: October 4, 2026

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1 Bookwork

1.1 Differential kinematics

Problem 1 Pseudo-inverse

[4]

- 1.1 For a redundant manipulator, which of the following expressions are well-defined at every robot configuration for any $\lambda > 0$? Select all that apply. The regularized expressions are known as “singularity robust” or “damped” inverses.

[1]

Gradescope:

- ☐ $(J^T J)^{-1} J^T$
- ☐ $J^T (J J^T)^{-1}$
- ☐ $(J^T J + \lambda I)^{-1} J^T$
- ☐ $J^T (J J^T + \lambda I)^{-1}$

- 1.2 Redundancy is not the only reason to damp. Consider a two-joint arm whose Jacobian at some configuration is

[3]

$$J = \begin{bmatrix} 1 & 0 \\ 0 & 0.001 \end{bmatrix} \quad (\text{meters per radian}),$$

It is square and the plain inverse exists. Now we ask for the end-effector velocity $V^d = (0, 1)^T$ m/s.

Gradescope:

- i. Using J^{-1} , the commanded joint velocity would be $\dot{q} = (0, \text{_____})$ rad/s.
- ii. Using the damped form $J^T (J J^T + \lambda I)^{-1}$ with $\lambda = 0.01$, it is $\dot{q} = (0, \text{_____})$ rad/s. (Four significant figures.)
- iii. That damped $\dot{q}$ is much smaller, but it does not achieve the requested end-effector velocity. The end-effector velocity it actually produces is $V = (0, \text{_____})$ m/s. (One significant figure.)

Going further: The damped gain applied to a singular value σ is $\sigma/(\sigma^2 + \lambda)$. For which value of σ is it largest, and how large does it get? Use that to say what a choice of λ is really choosing.

Problem 2 DiffIK via optimization

[5]

- 2.1 In class, we have seen that we can formulate the differential IK problem as a constrained optimization problem. What advantages does the optimization approach (OPT) have over using a singularity-robust pseudo-inverse (SRPI) on its own?

[1]

Gradescope:

- ☐ OPT produces reasonable robot velocities even near singularities whereas SRPI does not
- ☐ OPT will never cause the robot to stop before reaching its goal while SRPI can
- ☐ OPT can enforce the robot's actual joint limits and SRPI can't
- ☐ OPT can enforce the robot's actual bounds on joint velocities while SRPI can't
- ☐ OPT can enforce the robot's actual bounds on joint accelerations while SRPI can't

2.2 It is tempting to think you could get the same effect by computing the pseudo-inverse solution and then clipping it into the limits. Here is a two-joint arm with a one-dimensional task, at a configuration where

[4]

$$J = \begin{bmatrix} 1 & 2 \end{bmatrix}, \quad V^d = 1 \text{ m/s}, \quad |\dot{q}_i| \leq 0.3 \text{ rad/s for } i = 1, 2.$$

The optimization formulation is $\min_{\dot{q}} \|J\dot{q} - V^d\|^2$ subject to those bounds.

Gradescope:

- The (undamped) pseudo-inverse solution is $\dot{q} = (\text{_____}, \text{_____})$ rad/s, which is infeasible.
- Clip each component of that answer into $[-0.3, 0.3]$ and push it back through J . The end-effector velocity you get is _____ m/s.
- Solve the optimization instead. It returns $\dot{q} = (\text{_____}, \text{_____})$ rad/s, giving an end-effector velocity of _____ m/s.

Going further: The task here was one-dimensional, so both answers at least pointed the right way and only the speed was wrong. With a six-dimensional V^d and a cost of $\|J\dot{q} - V^d\|^2$, what can the QP's answer do that you might not want? What would you change so it can't?

Problem 3 Joint centering

[5]

Eq (23) in Chapter 3 of the notes describes how to do “joint centering” in the optimization formulation.

$$\min_{v_n} \|J^G(q)v_n - V^{Gd}\|_2^2 + \epsilon \|P(q)(v_n - K(q_0 - q))\|_2^2, \quad (23)$$

subject to constraints.

Here v_n is the next commanded joint velocity (equal to $\dot{q}$ for the fixed-base iiwa), and $0 < \epsilon \ll 1$ weights the joint-centering objective relative to end-effector tracking. The constraints are the joint position, velocity, and acceleration limits. The null-space projection is $P(q) = I - J^G(q)^+ J^G(q)$, where J^+ is the undamped Moore–Penrose pseudoinverse.

The “classic” formulation of a joint centering controller, not formulated using optimization, is:

$$\dot{q} = J^G(q)^+ V^{Gd} + P(q) K(q_0 - q).$$

3.1 Which of these are true for the iiwa?

[1]

Gradescope:

- ☐ $\dot{q}$ is a 6-d spatial velocity
- ☐ $\dot{q}$ is a 7-d joint velocity
- ☐ V^{Gd} is a 6-d spatial velocity
- ☐ V^{Gd} is a 7-d joint velocity
- ☐ V^{Gd} depends on the chosen representation for rotation
- ☐ V^{Gd} does not depend on the chosen representation for rotation
- ☐ $J^G(q)$ is 7×6
- ☐ $J^G(q)$ is 6×7
- ☐ $K(q_0 - q)$ has units of radians
- ☐ $K(q_0 - q)$ has units of radians/sec
- ☐ $P(q)$ is 7×6
- ☐ $P(q)$ is 6×7
- ☐ $P(q)$ is 6×6
- ☐ $P(q)$ is 7×7
- ☐ $J^G(q)(P(q)v) = 0$

$$\bigcirc J^G(q)(P(q)v) \neq 0$$

3.2 The classic controller is easy to understand for a two-joint arm with a one-dimensional task. Dropping the G superscript on J , take [4]

$$J = \begin{bmatrix} 1 & 1 \end{bmatrix}, \quad V^{Gd} = 0.2, \quad q = (0.6, 0), \quad q_0 = (0, 0), \quad K = 1.$$

Gradescope:

- The primary term is $J^+ V^{Gd} = (\text{_____}, \text{_____})$.
- The off-diagonal entry P_{12} of the projector is _____.
- The secondary term is $PK(q_0 - q) = (\text{_____}, \text{_____})$.
- Their sum is $\dot{q} = (\text{_____}, \text{_____})$, and $J\dot{q} = \text{_____}$ — unchanged, which is the point.
- Back on the real iiwa, away from singularities, the rank of $P(q)$ is _____.

Going further: Why would you want to do joint centering at all? And look again at the numbers above: the secondary term drives joint 1 back toward its center but drives joint 2 *away* from its center. Is it still doing its job? Then think about the iiwa, where you just found the nullspace is one-dimensional — and about what happens to that number when the arm reaches a singularity.

Problem 4 Hold my beer [5]

The differential IK QP is re-solved at every control tick, with timestep h , and its decision variable is the joint velocity $\dot{q}$ for this tick only. Limits that are naturally about position or acceleration therefore have to be rewritten as bounds on $\dot{q}$. Throughout, $h = 0.01$ s and we look at a single joint with upper limit $q^{\max} = 1.57$ rad and velocity limit $|\dot{q}| \leq 2.0$ rad/s.

4.1 A first-order guard against overshooting the position limit is to require that one timestep of $\dot{q}$ does not move you past it: $q + h\dot{q} \leq q^{\max}$. [3]

Gradescope:

- At $q = 1.50$ rad this bounds $\dot{q}$ above by _____ rad/s.
 - At $q = 1.56$ rad it bounds $\dot{q}$ above by _____ rad/s.
 - At $q = 1.50$, which gives the tighter upper bound on $\dot{q}$? ☐ position ☐ velocity
 - At $q = 1.56$, which gives the tighter upper bound on $\dot{q}$? ☐ position ☐ velocity
- 4.2 An acceleration limit $|\ddot{q}| \leq 10$ rad/s² becomes a bound relative to the velocity the previous tick actually commanded: $|\dot{q} - \dot{q}_{\text{prev}}| \leq h |\ddot{q}|^{\max}$. Suppose $\dot{q}_{\text{prev}} = 0.50$ rad/s and the cost alone would like $\dot{q} = 1.2$ rad/s. [2]

Gradescope:

- The acceleration limit confines $\dot{q}$ to the interval [_____, _____] rad/s.
- With q far from its limits, the QP therefore returns $\dot{q} = \text{_____}$ rad/s.

Going further: The velocity limit is a statement about this tick alone; the other two bring in q and $\dot{q}_{\text{prev}}$, so the controller is no longer memoryless. Two things can now go wrong that could not before. First, suppose the joint is running at 2.0 rad/s straight at $q^{\max}$: does the one-step position constraint above actually keep it safe, given the acceleration limit? Second, what happens to the QP if q somehow ends up slightly *outside* its limit? In each case, what would you change?

1.2 Motion planning

Problem 5 Search space [1]

Gradescope: Collision-free motion planning for robots involves searching for a path through the space of robot configurations. Which of the following are minimal representations sufficient for collision checking and goal specification under these assumptions? Assume the circular and cylindrical robots can translate

freely in the plane, and their heading does not affect collision checking or the goal. The cylinder stays upright at a fixed height. Assume the robot arms have fixed bases and all their joints are revolute; any joint limits are respected in the allowed ranges of the joint-angle coordinates.

- ☐ The (x, y) position of a circular robot moving in a plane.
- ☐ The (x, y) position of a cylindrical (holonomic) robot rolling on the floor of a 3D room.
- ☐ The (x, y) position of the center of an arbitrary rigid robot moving in a plane.
- ☐ The (x, y, θ) pose of an arbitrary rigid robot moving in a plane.
- ☐ The joint angles of a 6-dof robot arm.
- ☐ The pose (position, quaternion) of the end-effector of a 6-dof robot arm.
- ☐ The joint angles of a 7-dof robot arm.
- ☐ The pose (position, quaternion) of the end-effector of a 7-dof robot arm.

Going further: A 7-joint arm can change its elbow position while keeping the same end-effector pose. Why might this matter when planning around a shelf?

Problem 6 Dimension

[1]

Gradescope: Consider two Kuka iiwa 14 robots, each equipped with a WSG gripper, both rigidly mounted on a single cylindrical holonomic robot platform (1m tall, 1m diameter). Assume the platform moves on a flat floor and stays upright. Each gripper can open and close, and its two fingers are coupled to move symmetrically. What is the dimension of the minimal configuration space? Dimension: _____

Going further: Suppose both grippers are locked at fixed openings. Does this make collision-free planning necessarily easier, even though it reduces the configuration-space dimension?

Problem 7 Completeness and practicality

[6]

Gradescope: Consider the following motion planning domains (assume all obstacles are either polygons in 2D or polyhedra in 3D):

1. 2D Translating polygons
2. 2D Rotating/Translating polygons
3. 3D Translating (in a plane) polyhedra
4. 3D Translating polyhedra
5. 3D Rotating/Translating polyhedra
6. 2-joint 2D robot
7. 2-joint 3D robot
8. 7-joint 2D robot
9. 7-joint 3D robot

Answer one of *none*, *all* or *some* for each of the questions below.

For practicality, consider typical problems requiring useful geometric accuracy, rather than specially easy instances. Judge whether the method is generally computationally feasible in each domain.

7.1 Grid search is *complete* for which of these problem domains? _____ *Resolution complete?* [3]
 _____ *A practical choice?* _____

7.2 Classic PRM/RRT algorithms are *complete* for which of these problem domains? _____ [3]
 _____ *Probabilistically complete?* _____ *A practical choice?* _____

Going further: An RRT has run for a long time without finding a path. Can you conclude that no path exists? How might a narrow passage explain the result?

Problem 8 Cspace obstacles

[1]

Gradescope: Which of these represent valid approaches to computing the Cspace obstacle for translating convex polygons moving among convex polygonal obstacles? Apply each construction to one obstacle at a time. Take the Minkowski difference of A and B to be $A - B = \{a - b : a \in A, b \in B\}$.

- ☐ Convex hull of Minkowski difference of vertices of obstacle and robot
- ☐ Convex hull of Minkowski sum of vertices of robot and obstacle
- ☐ Star algorithm
- ☐ Union of copies of the robot with its reference point placed at every point of the obstacle, including its interior, keeping the robot's orientation fixed
- ☐ The same union using the “negated” robot, obtained by reflecting the robot through its reference point

Going further: For a circular translating robot whose reference point is its center, would forgetting to negate the robot change the Cspace obstacle? Why might testing only this shape hide a bug?

Problem 9 Differential drive

[1]

Gradescope: Consider a differential-drive (two wheel) rectangular robot. Select all that are true.

- ☐ This is a holonomic robot
- ☐ This is a non-holonomic robot
- ☐ Given two arbitrary (x, y, θ) poses in the plane (with no obstacles), this robot is not guaranteed to be able to move between them.
- ☐ By rotating in place, moving in a straight line and rotating again, this robot can reach any (x, y, θ) pose in the plane (with no obstacles).
- ☐ Even in the presence of obstacles, there is no advantage to moving backwards; going forwards is sufficient.

Going further: In an empty room, a differential-drive robot wants to end a short distance directly to its left with its original heading. Can it do so even though it cannot drive sideways? Describe a sequence of motions. How does this differ from a car?

Problem 10 Running times

[1]

Gradescope: Mark all that are true.

- ☐ The running time of complete motion planning algorithms grows exponentially (in the worst case) with the number of DOF of the robot.
- ☐ The running time of complete motion planning algorithms grows exponentially (in the worst case) with the number of objects in the environment.
- ☐ Probabilistically complete motion planning for a 7 DOF robot can typically be done in less than a milli-second, in the best implementations.
- ☐ Probabilistically complete motion planning for a 7 DOF robot typically takes tens of seconds, even in the best implementations.

Going further: Two planning problems use the same 7-DOF arm and the same number of obstacles. One has a wide route to the goal; the other has only a narrow passage. Why might their RRT running times be very different?

Problem 11 Grid-based planning

[4]

Let's consider grid-based motion planning. The idea is that we make a uniform grid in configuration space, and mark each cell as occupied if any part of it is not free. Then we can use A* to plan a path from the starting grid cell to the goal grid cell, stepping between *neighboring* cells that share a face or a

corner (so diagonal moves are allowed) and assigning a cost to each step that is the Euclidean distance between the two cells' centers. Read *finer* and *coarser* below as halving and doubling the cell size, so that the grids nest: every cell of a coarser grid is a block of whole cells of a finer one.

Gradescope:

11.1 Mark all that are true. [1]

- ☐ For any motion planning problem that has a solution with some clearance — a path keeping a positive distance from every obstacle — there is a grid granularity fine enough so you can find a path.
- ☐ If a path is found, it is guaranteed to be a shortest path in continuous Cspace, without restricting motion to the grid.
- ☐ If the search fails for some granularity then a solution might exist at a coarser granularity.
- ☐ If the search fails for some granularity then a solution might exist at a finer granularity.
- ☐ The number of cells grows exponentially in the number of dimensions of the cspace.
- ☐ The number of cells grows exponentially in the number N of divisions you make in each dimension of the cspace.

11.2 Now suppose one of the joints turns without limit, so that coordinate is an angle and its configuration space is a circle rather than an interval. We still divide it into N cells, now covering $[-\pi, \pi)$. For this joint, use the shortest angular difference when calculating step costs, including steps across the $-\pi/\pi$ boundary. Mark all that are true. [1]

- ☐ The grid has the same number of cells as before.
- ☐ The cell at the top of the range and the cell at the bottom of it have to be neighbors.
- ☐ If we leave those neighbors out, the planner can report failure at every granularity on a problem that has a solution.
- ☐ If we leave those neighbors out, the planner can return a path the robot cannot follow.
- ☐ A joint with limits, say $[-\pi/2, \pi/2]$, needs the same treatment.

11.3 A* also needs a heuristic. Let's use the straight-line Euclidean distance from a cell's center to the goal cell's center. For the first three choices, use a grid without wrap-around connections. Only the last choice uses the unlimited joint from the previous part. Mark all that are true. [1]

- ☐ It is admissible.
- ☐ It stays admissible however the obstacles are arranged.
- ☐ It stays admissible if instead every move to a neighboring cell, diagonal or not, costs 1, and the heuristic is measured in cell widths.
- ☐ It stays admissible for the unlimited joint of the previous part, if we go on computing the heuristic from the coordinates.

11.4 Take a 7-joint arm and divide each joint's range into $N = 100$ intervals. How many cells does the grid have? _____ [1]

Going further: Suppose we mark a cell free whenever its center is collision-free, rather than requiring the entire cell to be free. Could the planner now return a path that collides? Give a simple example.

2 Computational questions

2.1 Optimization and differential IK

Problem 12 Introduction to QP [5]

The purpose of this exercise is to get you familiar with the basics of what an instance of an optimization problem is, as well as how to solve it.

An optimization problem is usually written as

$$\begin{aligned} \min_x \quad & f(x) \\ \text{s.t.} \quad & g(x) \leq 0, \\ & h(x) = 0 \end{aligned}$$

We call x the **decision variable**, $f(x)$ the **cost function**, $g(x) \leq 0$ an **inequality constraint**, and $h(x) = 0$ an **equality constraint**. We usually denote the optimal solution by x^* . Most of the times, the constraints are hard-constraints, meaning that they must be fulfilled by the optimal solution.

Drake offers a very good interface to many solvers using `MathematicalProgram`. Let's try to solve a simple problem using `MathematicalProgram`:

$$\begin{aligned} \min_x \quad & x^2 \\ \text{s.t.} \quad & x \geq 3 \end{aligned}$$

Before we start coding, what do you expect the answer to be? You should persuade yourself that the optimal solution is $x^* = 3$, since that is value at which minimum cost is achieved without violating the constraint.

The code for this problem is in `ps3/intro_to_qp.py`.

- 12.1 Run `test1()`. You should see that we were successful in getting the expected solution of $x^* = 3$.
- 12.2 Now write `scalar_example_interior` to minimize $x^2 - 2x$ subject to $-1 \leq x \leq 3$. These bounds replace the previous constraint $x \geq 3$. Run `test2()` to observe the result.
- 12.3 A particular class of problems that we want to focus on this problem are [Quadratic Programs \(QP\)](#), which can be solved very efficiently in practice (even on the order of kHz). The general formulation of these problems are defined as follows.

$$\begin{aligned} \min_x \quad & \frac{1}{2}x^T \mathbf{Q}x + c^T x \\ \text{s.t.} \quad & \mathbf{A}x \leq b, \\ & \mathbf{A}'x = b' \end{aligned}$$

where $\mathbf{Q}$ is a positive-semidefinite, symmetric matrix. Note that the cost is a quadratic function of the decision variables, while the constraints are all linear. This is what defines a convex QP.

Here is a simple QP, already worked out for you:

$$\begin{aligned} \min_{x_0, x_1, x_2} \quad & x_0^2 + x_1^2 + x_2^2 \\ \text{s.t.} \quad & \begin{pmatrix} 2 & 3 & 1 \\ 5 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ & \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} \leq \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} \end{aligned}$$

This one is written for you as `vector_example` in the code; read it, then run `test3()` to see its solution. To conveniently write down constraints that are vector-valued, Drake offers `eq`, `le`, `ge` for elementwise constraints. It might take some time to learn the syntax of constraints. For a more in-depth introduction to `MathematicalProgram`, [Drake's MathematicalProgram tutorial](#) is very useful.

Now implement `solve_the_qp` to solve the following problem, and run `test4()`:

$$\begin{aligned} \min_{x_0, x_1, x_2} \quad & 2x_0^2 + x_1^2 + 4x_2^2 \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 2 & 3 \\ 2 & 7 & 4 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ & |x| \leq \begin{pmatrix} 0.35 \\ 0.35 \\ 0.35 \end{pmatrix} \end{aligned}$$

The last constraint means that each absolute value `abs(x[i])` is at most `BOX[i] = 0.35`. To express this using linear constraints, there are two routes:

- Break the constraints down to two constraints that don't involve the absolute value.
- Drake offers `AddBoundingBoxConstraint` which you may use in your implementation.

Be sure you understand both approaches.

12.4 Run `test4()`, which solves your program and checks the answer against the constraints.

[5]

Gradescope: The optimal solution is

$$x^* = (\text{_____}, \text{_____}, \text{_____}),$$

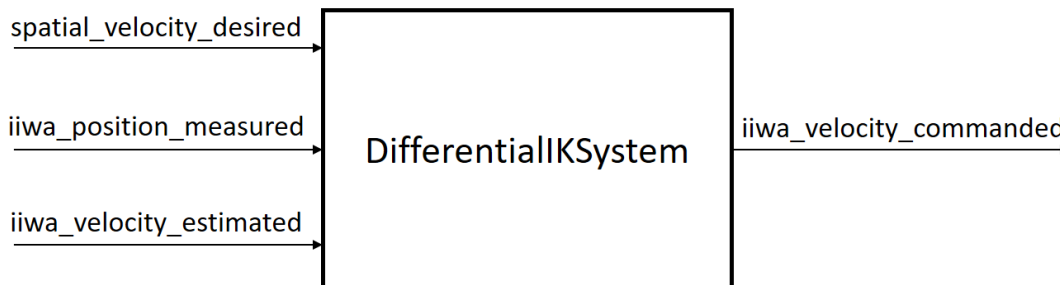
at which the cost is _____. (Give every entry to four decimal places.)

Problem 13 Differential IK as optimization

[8]

A differential IK controller converts a desired end-effector velocity into joint velocities. From Drake's system view of the world, how should we abstract this?

The controller is a `LeafSystem` with three input ports: `spatial_velocity`, `iiwa.position_measured`, and `iiwa.velocity_measured`. Its output port is `iiwa.velocity_command`. The measured position determines the Jacobian; the measured velocity is available to controllers that need it.



The diagram shows the conceptual signals; use the port names above when reading the supplied code.

Since we will go through several implementations of differential IK, `DifferentialIKSystem` takes a *function* that computes joint velocities from the Jacobian, the desired spatial velocity, and the rest of the robot's state. Swapping controllers is then a matter of passing a different function. `build_and_simulate` wraps the Drake plumbing around it:

- `HardwareStation`
- `DifferentialIKSystem`
- `ConstantVectorSource`, to command the desired velocity
- `Integrator`, to go from the controller's `iiwa.velocity_command` output to the station's `iiwa.position` input

All of this lives in `ps3/diffik_optimization.py`. Read the “code for students to be aware of” and “code for students to study” sections now.

13.1 Run `test1()`. It builds the whole diagram with `DiffIK_Zero`, a controller that commands zero velocity whatever it is asked for, so the manipulator should simply stay still.

13.2 In class we saw that to map a desired spatial velocity V^G to joint velocities we can use the relation [1]

$$V^G = J^G(q) v$$

and “invert” it with a Moore–Penrose pseudoinverse:

$$v = [J^G(q)]^+ V^G$$

That is `DiffIKPseudoInverse`, which we implemented in PS2. Run `test2()`, which asks the hand to move at 0.1 m/s in the $+x$ direction. (If Meshcat is slow to catch up, you can play back or scroll through the animation with its controls panel.)

Observe carefully what happens. Initially, the robot does move at 0.1 m/s in x . Verify that with respect to the colored world-frame axis (x is red).

Gradescope: Then what? Why? Submit one or two sentences in Gradescope.

13.3 How do we keep the benefits of the original method while improving the behavior near singularities? [1]
One idea from class is to hold to a **joint velocity limit**, which will at least stop the manipulator from doing something crazy. In the language of optimization, the problem we want to solve is

$$\begin{aligned} \min_v \quad & \|J(q)v - V_d\|_2^2 \\ \text{s.t.} \quad & |v| \leq v_{max} \end{aligned}$$

Implement `DiffIKQP`, transcribing and solving this program with `MathematicalProgram`, and run `test3()`.

Gradescope: What do you observe? Submit one or two sentences in Gradescope.

13.4 Run `test5()`, which calls `DiffIKQP` on the fixed data at the bottom of the file (`J_G.GIVEN` and the values beside it) rather than on a simulation, so that everyone’s answer is the same. It asks `DiffIKQP` for `V_G_DESIRED_FAST`, a command of 2 m/s in x and 1 m/s in y — far more than this arm can deliver from this configuration, so the velocity limit has to do some work. [4]

Gradescope: The first two joint velocities are (_____, _____); _____ of the seven joints end up at the 3.0 rad/s limit; and the largest joint speed the controller commands is $\max_i |v_i| = \text{_____}$ rad/s. (Four decimal places.)

13.5 In class we saw that on top of joint velocity constraints we can also constrain **joint positions** and **accelerations**, through a first-order Euler finite difference. Given v_{now} and q_{now} , we can ask for a v satisfying

$$\begin{aligned} \min_v \quad & \|J(q_{now})v - V_d\|_2^2 \\ \text{s.t.} \quad & |v| \leq v_{max} \\ & q_{min} \leq q_{now} + hv \leq q_{max} \\ & a_{min} \leq (v - v_{now})/h \leq a_{max} \end{aligned}$$

All of those constraints are written in joint space. But could we also end up with a linear constraint by imposing one in the *end-effector* space?

You must program a robot in a factory, and you would like a virtual wall that the robot is not allowed to put its hand through — space is tight and a human will be working there, or there are obstacles in the way.

Implement `DiffIKQP_Wall`, an optimization-based controller that minimizes $\|J(q_{now})v - V_d\|_2^2$ subject to two constraints:

1. the joint velocity constraint $|v| \leq v_{max}$, as before;
2. a constraint that the **position** of the end effector stay inside the box
 - `lower_bound = [-0.3, -1.0, 0.0]`
 - `upper_bound = [0.3, 1.0, 1.0]`

The current position of the hand is available as `p_now`, found by running forward kinematics. Constrain the gripper-frame origin in world coordinates using the last three Jacobian rows and prediction horizon $h = 0.004$ s. This horizon is not the controller's sampling period.

HINT: you can use a first-order approximation again, this time to ask where the hand will be next, given where it is now and how fast it is moving in end-effector space.

Run `test4()`, which draws the box and then commands a velocity that would carry the hand straight through it. Implemented correctly, the end effector stays inside the visualized box, almost as though the robot had a wall it could not cross. Small deviations can occur from linearization and simulation tracking errors. Note that constraining the *position* of the end effector is not the same as keeping the *entire* end effector inside the box — if part of it pokes out, that is fine. It is worth thinking about why.

13.6 Run `test6()`.

[2]

Gradescope: With your `DiffIKQP_Wall`, the hand's x velocity is _____ m/s and its predicted next x position is _____ m. (Four decimal places.) Use an absolute answer tolerance of 0.0001 in each quantity's units. The second value is a first-order prediction, not a measured position from the simulation.

2.2 Optimization-based inverse kinematics

Problem 14 Inverse kinematics for door opening

[11]

By using optimization-based inverse kinematics, we will solve a series of IK problems that can open a cupboard door. You can take a look at the [interactive_ik](#) notebook, which illustrates many of the concepts we go through in this question.

Our goal is to command the iiwa to open a cupboard door. We choose to open the door on the right. In `door_opening.py`, we provided you with setpoints ${}^W X^G(t)$ in the end-effector space, which gives us a nominal trajectory to reach and open the cupboard. “Nominal” refers to an ideal or baseline quantity. If you run `test1()` and open its Meshcat link, you will be able to visualize the trajectory. In the code, G is the gripper body frame, H is the handle frame, and ${}^W X_{\text{grasp}}^G = {}^W X^H {}^H X^G$, with ${}^H X^G = \mathbf{x_handleG}$. This trajectory divides the motion into the following segments:

- $0 \leq t \leq 5$: The end-effector linearly interpolates between the initial pose at $t = 0$, and the grasp pose required to grip the cylinder (${}^W \mathbf{X}_{\text{grasp}}^G$ at $t = 5$), while having the gripper open.
- $5 \leq t \leq 6$: The end-effector stays still at ${}^W \mathbf{X}_{\text{grasp}}^G$, and the gripper closes.
- $6 \leq t \leq 11$: The end-effector follows an arc of the handle as the door opens. The gripper remains closed.

We'll write code to convert this end effector trajectory to a joint-space trajectory using optimization!

The code for this problem is in `ps3/door_opening.py`.

14.1 Start by running `test1()` to see the robot, door, and the target door trajectory.

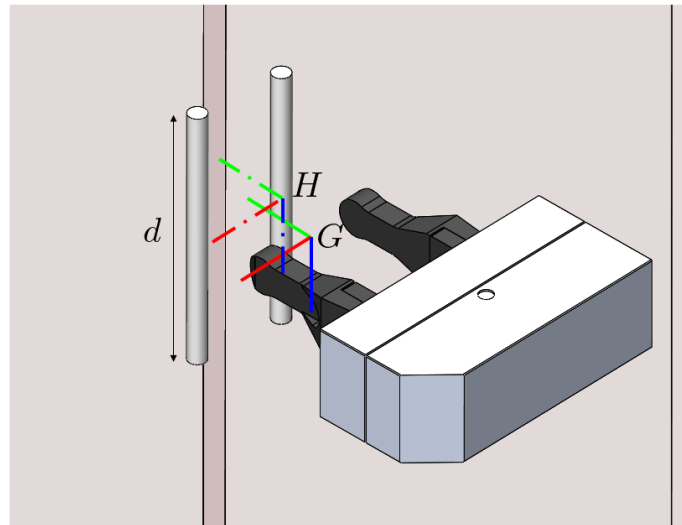
14.2 Just for this question, we'll focus on just the grasping pose, and disregard the rest of the trajectory. We define H as the frame at the center of the handle, as shown below, and denote our decision variables as q (joint angles of the arm), and the forward kinematics function as

[1]

$${}^W \mathbf{X}^G = f(q) \ .$$

We will write ${}^W \mathbf{R}^G = f_R(q)$ and ${}^W p^G = f_p(q)$ for the rotation and translation parts.

Recall one of the main ideas from the lecture: instead of having ${}^W \mathbf{X}^G = f(q)$ be constrained to exactly ${}^W \mathbf{X}_{\text{grasp}}^G$, the inverse kinematics problem can benefit by giving it freedom to pick from a set of values for ${}^W \mathbf{X}^G$ which makes sense for our problem.



Let us parametrize ${}^H X^G$ with six degrees of freedom: xyz positions and roll-pitch-yaw rotation of frame G with respect to frame H . Which three should we constrain to be exactly equal, and which three should we be willing to provide some slack? For this, let's ignore the length of the grippers. Select the three which we should constrain to be exactly equal to their nominal grasp values. For this part only, use an idealized contact frame G at the midpoint of two point-contact fingers, rather than the gripper body frame used in the code. The handle lies along H 's z axis, and the fingers close along G 's x axis. At the nominal grasp, $(x, y) = (0, 0)$ and $(\text{roll}, \text{pitch}, \text{yaw}) = (\pi, 0, 0)$, using $R = R_z(\text{yaw})R_y(\text{pitch})R_x(\text{roll})$. Consider small variations about this nominal grasp and ignore the handle's ends. (Remember the x-y-z axes are represented in red-green-blue colors respectively.)

Gradescope:

- ☐ x position
- ☐ y position
- ☐ z position
- ☐ x orientation (roll)
- ☐ y orientation (pitch)
- ☐ z orientation (yaw)

- 14.3 For the rest of this problem, we want to enforce equality constraints (up to a small tolerance) on both the position and the orientation. This is because we will be solving IK along an entire trajectory, not just for the grasp pose. [1]

We can write our final optimization along the entire trajectory. For each pose I , we solve:

$$\begin{aligned} \min_q \quad & \|q - q_{nom}\|^2 \\ \text{s.t.} \quad & f_p(q) = {}^W p^I \\ & f_R(q) = {}^W R^I \end{aligned}$$

Recall that q_{nom} are the baseline joint angles we define and hold fixed throughout the optimization. Here $q_{nom} = \mathbf{Q_NOMINAL}$; only the solver's initial guess changes between keyframes. What is the purpose of the quadratic cost term in our optimization formulation? Select all that apply.

Gradescope:

- ☐ To bias the solver toward a preferred “natural” configuration when multiple solutions exist.
- ☐ To ensure joint limits are respected.
- ☐ To guarantee that orientation constraints are satisfied exactly.

- ☐ To penalize large joint displacements from q_{nom} .

We're getting ready to implement the optimization. Before you get started, go through these exercises to get an idea of how to implement the constraints in Drake's `InverseKinematics` class.

- 14.4 Read the documentation for the `AddOrientationConstraint` method in Drake. [1]

Note that `AddOrientationConstraint` implements an inequality constraint, which is okay for this problem if we set the theta bound to be small.

What are frames $\bar{A}$ and $\bar{B}$, and what is their purpose? Select all that apply.

Gradescope:

- ☐ Arbitrary “virtual” frames that do not need to exist in the plant
- ☐ Frames existing in the `MultibodyPlant`
- ☐ Task frames constrained to have angle difference $\theta \leq \theta_{bound}$
- ☐ Reference frames such that ${}^{\bar{A}}R^A$ and ${}^{\bar{B}}R^B$ are constant throughout the optimization
- ☐ Reference frames such that ${}^{\bar{A}}R^A$ and ${}^{\bar{B}}R^B$ may change throughout the optimization
- ☐ They are the frames which we wish to align
- ☐ Use existing frames in the `MultibodyPlant` to express the task frames A and B in
- ☐ Allow you to define orientation constraints on arbitrary axes that are rigidly attached to bodies
- ☐ Ensure that you only give the orientation of A and B relative to the world and gripper, respectively

- 14.5 Read the documentation for the `AddPositionConstraint` method in Drake. Which of the following are true about the point Q , and what does the constraint enforce? [2]

Gradescope:

- ☐ It is a point fixed relative to frame B
- ☐ It is an arbitrary free-floating point in space
- ☐ It changes position relative to frame B during optimization
- ☐ We aim to constrain its position in frame A
- A. That the coordinates of Q , expressed in frame B , remain between ${}^A p^{Q-lower}$ and ${}^A p^{Q-upper}$
- B. That the distance between A and B is fixed
- C. That the coordinates of Q , expressed in frame A , remain between ${}^A p^{Q-lower}$ and ${}^A p^{Q-upper}$

Time to implement!

- 14.6 You will write two functions, using Drake's `inverse_kinematics` package.

First, `add_pose_constraint`, which asks an IK program for one frame to sit at one pose. Implement constraints on `ik` using `AddPositionConstraint` and `AddOrientationConstraint`. Note that these implement inequality constraints, which is okay if our error tolerance is small.

This one is deliberately written in terms of a frame, a pose and two tolerances, with nothing in it about the door: *the later IK problem in this pset imports it from here*, so the version you write now is the one that will be constraining the mobile manipulator. Keep it general.

Second, `create_q_knots`, which accepts a list of key frames `pose_lst` and converts them to joint coordinates. For each keyframe i :

- Call `add_pose_constraint` to put the gripper on that keyframe.
- Add a joint-centering cost on `Q_NOMINAL`.
- If `i==0`, set the initial guess to be `Q_NOMINAL` using `prog.SetInitialGuess`. Otherwise, set the initial guess to be the solution you obtained on the previous IK problem.

Note: When we define our constraints, we want to enforce the following tolerances

- Our position tolerance should be such that end effector matches the desired pose to within 1mm translation along each axis

- Our orientation tolerance should be such that the rotation is off by no more than $0.01 \cdot \pi$ radians

When you have implemented your function, run `test2()`. It smoothly interpolates (using a cubic spline—read about this if you haven’t encountered the idea before) between the key-point joint configurations you found, and then executes it on the robot. Pose tolerances apply at the keyframes; joint-space interpolation need not maintain them between keyframes.

You should now see the robot following the trajectory to open the door!

- 14.7 **Gradescope:** When it looks good, run `test3()` and enter the returned joint angles for the final pose in the trajectory, to 4 decimal places. (The last 2 dimensions of q are the gripper joints, so we can ignore those.) Be sure you are using the proper tolerances for the orientation and position constraints ($0.01 \cdot \pi$ for orientation and 0.001 for position). The supplied `check_final_configuration` checks whether a returned seven-angle vector meets the final pose and joint limits; equivalent feasible answers need not match one reference vector.

[6]

Problem 15 Inverse kinematics for a mobile manipulator

[6]

In this problem, we’ll solve inverse kinematics for one arm on a PR2 robot. This robot model has been modified to have convex collision geometries, and to simplify this problem, we have welded (fixed) several joints that are irrelevant to the kinematics of the arms.

The code for this problem is in `ps3/mobile_base_ik.py`.

- 15.1 Given a `RigidTransform X_WG`, we want to compute a robot configuration placing the left gripper at that pose. We use optimization to solve the IK problem, and we repeatedly solve the program with random initializations within a fixed attempt limit.

`solve_ik` is written for you: it builds the scene, sets up the `InverseKinematics` program with a cost that keeps the configuration near a nominal configuration, and runs the restart loop. It seeds NumPy’s random number generator with `IK_SEED` before the first restart, so that a run repeats on your own machine; leave the seed alone. You write the two procedures it calls.

Implement `add_ik_constraints`, which adds the constraints the configuration has to satisfy:

- Put the gripper frame on the desired pose. You wrote this in the door-opening problem, and it is imported from there — call your `add_pose_constraint` rather than writing the constraints again.
 - The end effector should match the desired pose to within 1mm translation along each axis
 - The rotation should be off by no more than 1 degree. Note that this is a different tolerance from the door-opening problem’s, which is why `add_pose_constraint` takes them as arguments.
- If `fix_base` is True, constrain the base pose (x, y, θ) to be equal to `base_pose`.
- Add a collision free constraint with `AddMinimumDistanceLowerBoundConstraint`. The minimum distance between any pair of collision geometries considered by Drake after collision filtering should be at least 1cm.

Implement `random_initial_guess`, which returns one configuration for the restart loop to start the solver from: draw each joint value uniformly within the robot’s joint limits. The plant reports those bounds as arrays with one number per position, in the same order as the configuration vector:

- `plant.GetPositionLowerLimits()`
- `plant.GetPositionUpperLimits()`

Some of the positions — the base’s x , y and θ — have no limit, and come back as $\pm\infty$; use the range $[-\pi, \pi]$ for those instead.

Run `test1()`, which fixes the base and runs all twenty restarts instead of stopping at the first success. It draws each configuration the solver stopped at — solved or not — and waits for you to press Enter, then reports how many succeeded and the cost of each solution.

- 15.2 The target is reachable from this base pose, but not every restart finds it. Roughly how many of the twenty succeed?

[1]

Gradescope:

- A. None.
- B. Fewer than one in ten.
- C. A substantial fraction, well short of all.
- D. All twenty.

15.3 `test1()` prints the cost of each solution, and they differ. What does the cost measure? [1]

Gradescope:

- A. How far the gripper is from the target pose.
- B. How long the solver took.
- C. How far the configuration is from the nominal configuration.
- D. How close the robot came to colliding.

15.4 In `test1()`, where the base is fixed, what makes this case hard? Pick all that apply. [1]

Gradescope:

- ☐ It has 15 degrees of freedom
- ☐ It has 7 degrees of freedom
- ☐ One of the arms is sticking out straight and can't be moved
- ☐ It has to avoid the shelves

15.5 `test2()` tries four target positions, first from `test1()`'s base pose and then with the base free to drive. Which one can the arm reach only if the base moves? A failed IK search does not prove unreachability; use the robot's reach and joint limits to interpret the results. [1]

Gradescope:

- A. $(-0.83, 0.18, 1.40)$
- B. $(-0.83, 0.50, 1.40)$
- C. $(-0.83, 1.30, 1.40)$
- D. $(-0.83, 0.18, 2.30)$

15.6 Which one is out of reach wherever the base goes? [1]

Gradescope:

- A. $(-0.83, 0.18, 1.40)$
- B. $(-0.83, 0.50, 1.40)$
- C. $(-0.83, 1.30, 1.40)$
- D. $(-0.83, 0.18, 2.30)$

15.7 If we vary the x and y coordinates of the base pose for the fixed base case, we notice that sometimes IK fails to find a feasible solution. Which of the following base poses leads to no solution being found within the allowed attempts in your run? Use `test3()` to figure this out. [1]

Gradescope:

- A. $[-1.23, 0.05, 0]$
- B. $[-1.5, -0.2, 0]$
- C. $[-1.8, 0.2, 0]$
- D. $[-1.4, 0.3, 0]$

2.3 Motion planning

Problem 16 RRT planning

In our examples so far, we were able to get away with explicitly defining poses of interest and interpolating [6]

between them without any consideration of our environment. Unfortunately, with this approach, navigating cluttered environments while avoiding both collisions and system limitations is a major challenge. To address this, we'll use motion planning.

In this problem, we'll build two sampling-based motion planners. We'll start with the famous Rapidly-exploring Random Tree (RRT) algorithm, then extend it to the popular two-tree variant, RRT-Connect.

Optional background reading [Kuffner & LaValle, "RRT-Connect: An Efficient Approach to Single-Query Path Planning" \(ICRA 2000\)](#).

The code for this problem is in `ps3/rrt_planning.py`.

- 16.1 Study the code in the **Code for students to study** section:

`iiwa_configuration_space`, `PlanningProblem`, and the `RRT_tools` class. Your planners are written in terms of these; be sure you understand each method before you use it.

- 16.2 Run `test1()` to see the problem instance. It shows the arm at q_{start} , pauses, and then shows it at q_{goal} , with the goal gripper pose drawn as a triad.

- 16.3 Implement the RRT algorithm, `rrt_planning` by implementing and using the helper `extend`. The path should reach the goal and pass the provided `edge_is_free` check on every edge.

For `extend`, the pieces are all in `RRT_tools`: `nearest_node`, `step_toward`, `edge_is_free` and `add_node`, and it reports one of the three `Status` values. For `rrt_planning` you also need `new_tree`, `sample_configuration` and `path_from_root`; the start and goal are `problem.start` and `problem.goal`, and `prob_sample_q_goal` is how often the goal itself is the sample.

Run `test2()`, which plans and then plays the path back in Meshcat. Do you notice the "RRT Dance"? Run it several times. Roughly how often does it succeed? Don't panic if it sometimes fails. Feel free to experiment with `max_iterations` and `prob_sample_q_goal`.

This simulation illustrates the limits of sampled planning checks. Contact can result from an inter-sample collision or from tracking error during execution. Compare any contact you observe with the finer recheck in part 16.6; passing that recheck still does not prove continuous collision freedom. A run with large contact forces or objects knocked over is not a successful execution: inspect the path and replan.

- 16.4 RRT-Connect builds on the basic RRT algorithm by growing two trees simultaneously: one rooted at the start and the other at the goal.

Implement the RRT-Connect algorithm, `rrt_connect_planning`, and run `test3()`. It plans and plays the path back the same way `test2()` does.

Reuse your `extend`. After successfully extending the first tree, repeatedly extend the second tree toward the new node until `Reached` or `Trapped`. One `tools` object serves both trees; a node's configuration is `node.value`, which is what you extend the other tree toward. When the trees meet, `path_from_root` on each side gives a path from that tree's root to the meeting point, and `join_paths` splices them once the goal tree's path is turned around.

- 16.5 Run `test4()`, which runs both planners several times and prints the outer iterations and planning time for each trial. An RRT-Connect iteration may contain several extensions, so compare times as well as iteration counts.

- 16.6 **Gradescope:** `test2()` and `test3()` re-check the path they were given with `first_collision`, which walks it at a maximum per-joint spacing of `CHECK_STEP` — a tenth of the `EDGE_STEP` the planners used. Paths the planner accepted are sometimes reported `IN COLLISION` by this finer check. Explain why that happens, and say whether checking every `CHECK_STEP` instead would guarantee that the entire continuous path is collision-free. [1]

- 16.7 Consider the following table, which indicates approximately how much, when the arm is fully outstretched, each joint would have to move (with the other joints fixed), to cause a 1 mm motion in the fingertip. The arm is at $q = 0$, pointing straight up, and the fingertip is the end of one finger pad, 6 cm off the arm's axis. What explains the pattern of the magnitudes of these numbers? [1]

joint	rotation (rad)	rotation (deg)
1	0.0167	0.955
2	0.00092	0.053
3	0.0167	0.955
4	0.00145	0.083
5	0.0167	0.955
6	0.00347	0.199
7	0.0167	0.955

Gradescope: Mark all that are true:

- ☐ These values argue that we should use different spacing for different joints when interpolating in the RRT.
- ☐ This same pattern of magnitudes will be the same no matter what configuration the robot is in.
- ☐ This is the Jacobian.
- ☐ This is the inverse Jacobian.

16.8 Consider the case where we let our RRT algorithm run forever, i.e. `max_iterations` grows without bound. If there is no path to the goal, will RRT warn you? If there is a path to the goal, how does the probability of RRT failing to find a path behave as the number of iterations goes to ∞ ? Assume a feasible path has positive clearance, sampling covers the free space, and local connections are checked correctly. [1]

Gradescope:

- A. Yes / It approaches zero
- B. No / It approaches zero
- C. Yes / It remains greater than zero
- D. No / It remains greater than zero

16.9 **Gradescope:** Upload the video of your robot following the RRT-Connect trajectory as an mp4 to Gradescope. Record the complete `test3()` playback with the scene visible. After the simulation finishes, use Meshcat's playback controls to return to the beginning. Start a screen recorder on that browser region, play the recording through once, and stop the screen recorder. Save or export as MP4; renaming another video format to .mp4 does not convert it. Open the exported file to verify that the complete motion is visible. The course file-size limit is 500 MB. [3]

2.4 Additional problems for 6.4212

Problem 17 Exact edge checking 6.4212 only

Our RRT decides whether a straight edge in configuration space is free by testing configurations at most `EDGE_STEP` apart in each joint, and the RRT problem ended with the observation that no fixed spacing *proves* an edge free. Schwarzer, Saha and Latombe ("Exact collision checking of robot paths") point out that a collision checker that reports a *distance*, rather than a yes or a no, can do better.

Consider one robot among static obstacles, and define two quantities. Here, collision-free refers to robot-obstacle collisions.

- The **clearance** $\eta(q)$ is the signed workspace distance between the robot, at configuration q , and the nearest obstacle. It is zero at contact and negative during penetration. The robot is collision-free at q exactly when $\eta(q) > 0$.
- A **motion bound** is a constant $\rho > 0$ such that, when the configuration moves along the straight line from any q to any q' , no point of the robot travels farther than $\rho \|q - q'\|_2$ in the workspace.

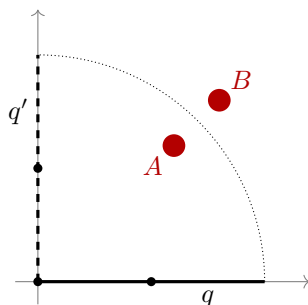
Theorem. For endpoints with $\eta(q) > 0$ and $\eta(q') > 0$, if

$$\rho \|q - q'\|_2 < \eta(q) + \eta(q'),$$

then every configuration on the segment from q to q' is collision-free.

First evaluate both endpoint clearances and reject either endpoint if its clearance is nonpositive. When the test fails it says nothing either way, so the edge checker *bisects*: it computes η at the midpoint; if that is not positive it reports a collision; otherwise it applies the test to each half, recursively. The edge is free once every piece has passed.

- 17.1 Prove the theorem. (Suppose some point p of the robot touches an obstacle partway along the segment. How far has p travelled by then, and how far does it still have to go?)
- 17.2 A planar arm has two zero-thickness links of length 1 and configuration $q = (\theta_1, \theta_2)$, with θ_2 measured relative to link 1. Show that along a straight line in configuration space no point of the arm travels farther than $2|\Delta\theta_1| + |\Delta\theta_2|$, and conclude that $\rho = \sqrt{5}$ is a motion bound for this arm.
- 17.3 The arm sweeps from $q = (0, 0)$ to $q' = (\pi/2, 0)$: it stays straight and turns a quarter circle. The only obstacle is a round post of radius 0.1, which stands either at $A = (1.2, 1.2)$ or at $B = (1.6, 1.6)$. The clearances along the sweep are



θ_1	0°	22.5°	45°	67.5°	90°
η , post at A	1.100	0.549	-0.100	0.549	1.100
η , post at B	1.500	0.771	0.163	0.771	1.500

With the post at A, a fixed-step checker tests $\theta_1 = 0^\circ, 10^\circ, \dots, 90^\circ$. What does it report, and is it right?

- 17.4 With the post at A, what does the bisecting checker do?
- 17.5 With the post at B, at which configurations does the bisecting checker evaluate η , and what does it conclude?
- 17.6 As described, the bisection can run forever in exact arithmetic. Describe a segment on which it does. Then suppose the checker instead rejects any sampled configuration with $\eta \leq \delta$, for a fixed margin $\delta > 0$: show that it never subdivides a piece shorter than $2\delta/\rho$, and conclude that checking at a *fixed* spacing ε is sound provided every sample has clearance greater than $\rho\varepsilon/2$. Use Euclidean configuration-space distance and sample both endpoints.

Problem 18 Exact edge checking for the iiwa 6.4212 only

Now put the bisecting checker of the previous problem into the RRT. The code is `exact_edge_check.py`; it imports the scene, the planning problem and your `rrt_connect_planning` from `rrt_planning.py`.

- 18.1 Study `joint_radii` and `ClearanceChecker`. `joint_radii` returns, for each joint j , a bound r_j on the distance from that joint's axis to any point of the arm or gripper that the joint moves; it is the iiwa's version of the (2, 1) of the planar arm. `ClearanceChecker.clearance` is $\eta(q)$, from Drake's `ComputeSignedDistancePairwiseClosestPoints`. Implement `motion_bound`, which turns the r_j into ρ , and run `test1()`.
- 18.2 Implement `is_certified`, which is the theorem, and `exact_edge_is_free`, which is the bisection with the margin `DELTA`. Compute η once per configuration: pass the clearances at the two ends of a piece down the recursion with it. Also ask `collide` about each configuration you test; the last part explains why. `DELTA` is a rejection threshold at sampled configurations; the certificate guarantees positive clearance between samples, not a `DELTA` margin. Run `test2()`. It plans with the fixed-step planner, as before, and then puts each edge of the path through your check.

- 18.3 Run `test3()`, which plans with your check inside RRT-Connect, by way of `ExactPlanningProblem`. How often do these paths fail the `CHECK_STEP` re-check? Why? Consider robot–obstacle collisions and self-collisions separately.
- 18.4 The last line `test1()` prints is the margin that would make the planner’s fixed `EDGE_STEP` sound for robot–obstacle collisions, by the last part of the previous problem. Compare it with the clearance at q_{goal} . What would happen if we made the fixed-step planner sound by requiring that margin of every sample?
- 18.5 `clearance` leaves out pairs of the arm’s own links, and self-collision is caught only by the calls to `collide`, at the configurations that happen to be tested. The theorem does hold for such a pair, with the same ρ , if the motion of the descendant link is measured in the frame of the ancestor link. But at q_{start} the collision boxes of links 5 and 7 are 1.9 cm apart. What would including that pair in η do to the checker? What better bound is available for it?

Upload one 1–2 page PDF with your answers to Questions 17 and 18 (no code) to Gradescope. This combined submission is worth 11 points.

3 What you should understand about the code you just produced

- In part 13.5 which constraints are active in the solution?
- In part 13.3 what would be another solution to the crazy behavior?
- Why does adding λI change the behavior of the pseudo-inverse near a singularity, and what does the robot actually do differently?
- When the QP for differential IK becomes infeasible, what should the controller do?
- What is the cost function doing in the door-opening IK, and what would change if you deleted it entirely?
- Why does the mobile base make the IK problem easier rather than harder, given that it adds decision variables?
- RRT-Connect grows two trees instead of one. Where does the speedup come from?
- What is the difference between a circular interpolation and a Piecewise Pose interpolation for the gripper in the door-opening example?
- When we use optimization-based IK on a whole trajectory, how do we initialize each solver call? Why?
- Does mobile-manipulation IK seem to be easier when the whole base is free to move or when it is fixed in a known feasible configuration? Why?
- In the RRT code, we have `EDGE_STEP = np.radians(1.0)`. This could cause a problem—would it be that we return a path that is actually not collision free, or that we fail to find a collision-free path if one exists? Give an example situation where this problem could arise.
- In RRT, what would happen if you set `GOAL_BIAS` to 0? to 1?
- How would the interpolation in the RRT method have to change if we had a joint that rotates freely all the way around?